

## **LAMINECTOMY**

**Lamina:** Vertebral spinous processes → have you ever felt the bumpy bones on the back of somebody? It's not the body of the vertebrae, it's the wingy thingies → those are the lamina.

Do you remove the round bodies of them? NO. You removed the wings. The posterior processes of the vertebral bones.

**WHY a laminectomy?** To remove nerve root compression because sometimes when they come out of the spinal cord there's calcium or herniated disks or inflammatory masses pressing on them → cut away the bone → give nerve more room to exit → relieve compression of nerve root

### **S&S of Nerve Root Compression**

#### **Three P's:**

1. Pain
2. Paresthesia - numbness and tingling
3. Paresis - muscle weakness

**Most important thing to pay attention to in any nerve question: LOCATION!**

- Location determines the symptoms
- Location determines the prognosis
- Location determines the treatment

### **THREE LOCATIONS OF A LAMINECTOMY**

1. Cervical → neck
2. Thoracic → upper back
3. Lumbar → lower back

**Most important pre-op ax for a CERVICAL laminectomy?**

*Cervical innervates → diaphragm and arms*

Therefore, assessment of:

1. Breathing
2. Back up answer → function of arms and hands (back up answer)

**Most important pre-op ax for a THORACIC laminectomy?**

*Thoracic innervates → abdominal muscles and gut muscles*

Therefore, assessment of:

1. How a patient coughs
  - a. Put hand on abdomen, cover your mouth, and on my signal i want you to cough. You should feel a contraction → You CAN cough.
  - b. With a thoracic spinal cord injury. You can no longer contract your abdomen. You are paralyzed → You CANNOT cough. You can breathe.
2. Back up answer → Bowel sounds

**Most important pre-op ax for a LUMBAR laminectomy?**

*Lumbar innervates bladder and legs.*

Therefore, assessment of:

1. Last time of void
  - a. Is it empty or full?
2. Back up answer → function of legs

## POST OP LAMINECTOMY

**#1 answer → LOG ROLL. Always pick log roll if it's there!**

### Mobilizing Patients After Laminectomy:

- Do not dangle
  - No sitting at side of bed
- Go from lying down immediately to walking around
- Can they sit on the side of the bed to overcome orthostatic hypotension? Yes.
  - They should not sit with feet on the ground/feet off the ground/sitting at side of bed → NO!
- Do not sit longer for 30 minutes
  - **Question:** What typical post op order would you question with laminectomy? Up in chair for 2 hours TID.
    - Up in meals is okay because meals are only 30 minutes
- They may walk, stand, lie down WITHOUT restrictions
  - RESTRICTIONS ONLY WITH SITTING
- What worker in hospital has the highest risk of back injuries?
  - Unit clerk because they sit all day

### Post-op Complications

Depends on location!

- **Cervical:** They won't breathe deeply after surgery → PNEUMONIA
- **Thoracic:** They won't cough so well → PNEUMONIA and ILEUS
- **Lumbar:** URINARY RETENTION followed by PROBLEMS WITH LEGS

**Question:** You are caring for patient with lumbar oligodendrocytoma. What is the number one problem?

- a. Airway
- b. Ileus
- c. Cardiac arrhythmia
- d. **Urinary retention**

→ Because it's LUMBAR. All you need to know is the location, you don't need to necessarily know what it is!

- Typically you don't have chest tubes with laminectomies but with an **anterior thoracic laminectomy** → from the front, you go through the chest, to the spine.
  - Pneumothorax → requiring CT
- **Laminectomy WITH FUSION** → takes a bone graft from the iliac crest
  - If you take the disc out, you cannot have bone on bone
  - You take bone from hip, and put in between to fuse it so there's no grinding
  - Therefore they will have TWO incisions
    - One on hip and one on spine
      - Of those two incisions, which one will have more pain? THE HIP.
      - Which has the most bleeding and drainage? THE HIP.
      - Which has the highest risk for infection? THEY ARE EQUAL.

- Highest risk for rejection? THE SPINE.
- Therefore, HIP causes more problems.
- Why do surgeons want to get rid of hip incision? Shorter recovery, decreases RF infection, decreases pain.
- Surgeons are using cadaver bones from bone banks to just have one incision instead of two
- Nowadays they are moving away from cadavers to synthetic bonding materials

**DISCHARGE TEACHING:** Four temporary restrictions and three permanent ones

**TEMPORARY:**

1. Do not sit for longer than 30 minutes which applies for 6 weeks.
  2. Lie flat and log roll for 6 weeks.
  3. No driving for 6 weeks
  4. Do not lift more than 5 pounds for 6 weeks.
- a. A gallon of milk is about 5 pounds.

**PERMANENT:**

1. Do not pick up objects by bending at the waist; bend with the knees!
  2. Cervical laminectomies are not allowed to lift anything OVER THEIR HEAD.
- a. Need a step stool in the house
3. No jerky car rides, bike rides, roller coasters, etc.

This lecture can be used to get any spinal cord questions correct!

## **LAB VALUES**

Know which lab values are more dangerous than what other lab value? Prioritization.

### **PRIORITY LEVELS**

- A. Low priority: Yes it's abnormal, may be presence of disease, but no big deal. For instance, you could wait all night then have it assessed tomorrow, it's okay.
- B. Still a low priority: It is abnormal, you need to be concerned, you just need to watch them closer.
- C. High priority: It is critical; you must do something about it.
- D. Highest priority

| DESCRIPTION                                                           | NORMAL LVLS | PRIORITY                                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Serum Creatinine                                                      | 0.6 – 1.2   | A<br><br>Phone call worthy only if they were going for a test that had a dye in it                                                                                                                                                                |
| INR<br>- Monitors Coumadin therapy<br>- Variation of prothrombin time | 2's to 3's  | 4 and above = C<br><br><b>What do you do?</b><br>Follow an order<br><br>First thing you do is HOLD<br><br>After you hold, then you assess but a FOCUSED ax based on which body part<br><br>Prepare to give the drug<br><br>Call physician/RT/etc. |
| Potassium                                                             | 3.5 – 5.3   | A LOW potassium is a C<br>Do you hold? NO.<br>Ax the heart.<br>Prepare to administer K+<br>Call MD<br><br>Between 5.4 – 5.9 is a C<br>Hold all K+<br>Ax the heart<br>Prepare to give kayexelate and D5W and regular insulin<br>Call MD            |

|                 |                                                      |                                                                                                                                                                                                                                                                                                                     |
|-----------------|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 |                                                      | 6 and great is a D<br>Deadly dangerous<br>Same protocol but more staff on hands<br>Don't leave your pt and do what you can do at the bedside                                                                                                                                                                        |
| pH              | 7.35 – 7.45                                          | pH in 6's is a D<br>Ax vitals b/c pH pt is going down!<br>Only way to correct is to treat the underlying cause<br>Low pH get the physician in on the case!!!<br>Stay with the patient                                                                                                                               |
| BUN             | 8 – 25<br><br>How many bun's do you buy in a pack 8! | <8 is A priority → ax for dehydration                                                                                                                                                                                                                                                                               |
| HgB             | 12 – 18                                              | 8 – 11 it's a B priority → ax for low HgB → bleeding or malnutrition (anemia)<br><br>If < 8 it's a C → do something<br>Ax for bleeding<br>Prepare to administer blood<br>Call the MD                                                                                                                                |
| Bicarb          | 22 – 26                                              | Abnormal bicarb is an A                                                                                                                                                                                                                                                                                             |
| CO2             | 35 – 45                                              | A CO2 that's high but in the 50's → those are C levels<br>It's critical<br>Not COPDers just normal<br>Ax respiratory status<br>Pursed lip breathing<br>Call MD<br><br>A CO2 that's in the 60's → D<br>Super high priority<br>Do not leave the room<br>Prepare to intubate and ventilate<br>Call RT and then call MD |
| Hematocrit      | 36 – 54<br><br>* It is the Hgb x 3                   | Elevated Hct is a B<br>Ax dehydration                                                                                                                                                                                                                                                                               |
| pO2 (blood gas) | 78 – 100                                             | If it is low but still in 70's (70-77) it is a C<br>Ax respiratory<br>Prepare to administer O2<br><br>When hypoxic, the heart will increase FIRST then respirations will                                                                                                                                            |

|                        |                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        |                                                                                                                   | <p>Two common causes for episodic tachycardia:</p> <ul style="list-style-type: none"> <li>- Hypoxia</li> <li>- Dehydration</li> </ul> <p>In 60's it's a D</p> <p>Respiratory failure</p> <ul style="list-style-type: none"> <li>- CO2 in 60's</li> <li>- O2 in 60's</li> <li>- Both in 60's → intubate and ventilate</li> </ul> <p>Throw on O2 to make comfortable</p> <p>Ax pt</p> <p>Prepare to intubate or ventilate</p> <p>Stay with patient</p> <p>Call MD</p> |
| O2 SAT                 | 93 – 100                                                                                                          | <p>Anything less than 93 is a C</p> <p>Ax resp</p> <p>Prepare O2</p> <p>In pediatrics → &lt; 95% is a C</p> <p>COPDers run high on CO2 and low on O2</p> <p>Anemia falsely elevates SaO2; therefore look for other ways to ax oxygenation</p> <p>dye in last 48 hours also falsely elevates and invalidates O2 because dye colours the blood</p>                                                                                                                    |
| BNP<br>Best ind of CHF | < 100                                                                                                             | <p>📄 100 is a B → CHF but you're not gonna die; it's a chronic condition</p>                                                                                                                                                                                                                                                                                                                                                                                        |
| Na                     | 135 – 145                                                                                                         | <p>Abn is a B</p> <p>If high ax for dehydration</p> <p>If low ax for overload</p> <p>If question says Na is abn and change in LOC it is a C → safety issue</p>                                                                                                                                                                                                                                                                                                      |
| WBC                    | <p>Total WBC 5,000 – 11,000</p> <p>Absolute Neutrophil Count (ANC)</p> <p>📄 500</p> <p>CD4 count</p> <p>📄 200</p> | <p>All abnormal values are C</p> <p>Ax for signs of infection</p> <p>Place them on <b>neutropenic precautions:</b></p> <ul style="list-style-type: none"> <li>✓ Follow strict hand washing</li> <li>✓ Shower BID with antimicrobial soap</li> <li>✓ Avoid crowds</li> <li>✓ Private room</li> <li>✓ No fresh flowers or potted plants</li> <li>✓ Low bacteria diet- no raw fruits/veggies, no undercooked meat.</li> </ul>                                          |

|           |                          |                                                                                                                                                     |
|-----------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
|           | When < 200 → AIDS!       | ✓ No water drinking if its been sitting for longer then 15 minutes.<br>✓ Vitals Q4H<br>✓ Check WBC daily<br>✓ Avoid reusable plates/silverware etc, |
| Platelets | Thrombocytopenic <90,000 | Thrombocytopenic is a C → bleeding precautions<br><br>If < 40,000 → D<br>Prepare transfusion and call doctor                                        |
| RBC       | 4 – 6 million            | Abnormal is a B                                                                                                                                     |

If they give you an elevated blood value, you have no idea, dehydration is a good guess because when you're dehydrated, your blood cells are inc = dehydration.

- Ax before you do unless delaying what you do before you ax would put your patient at higher risk.
- Delaying stopping the blood to do an ax → inc risk; therefore you do before you assess
- Position first if there's TWO DO'S
- But what if it's a FIRST question versus BEST question?
  - BEST = O2
  - FIRST = Raise HOB

### **MEMORIZE THE 5 D'S**

1. pH in the 6's
2. Potassium in the 6's
3. CO2 in the 60's
4. PO2 in the 60's
5. Platelets <40,000

### **THEN, MEMORIZE ALL THE C'S**

All the rest, you don't have to remember since they're not important!

## **MATERNAL NEWBORN OVERVIEW**

### **PREGNANCY DUE DATE**

1. Take first day of the last menstrual period
2. Add 7 days
3. Subtract 3 months

#### **Example:**

Her last menstrual period was June 10 – June 15

June 10 + 7 days = June 17

June 17 – 3 months = March 17 of the following year

Note: June is the 6<sup>th</sup> month – 3 = 3<sup>rd</sup> month which is March

### **WEIGHT GAIN**

1. Average typical pregnancy is what they are talking about here.

Total weight gain: 28 lbs +/- 3.

2. In the first trimester, she gains 1 pound, each month.

First trimester is 3 months

She gains a total of 3 pounds in 3 months

3. In the second and third trimester, she is gaining much faster. She gains 1 pound, per week.

On NCLEX, if they give you a woman in a particular week of gestation, you have to be able to predict what her weight gain should be.

### **IDEAL WEIGHT GAIN CHART**

| <b>Week</b>                                          | <b>Weight Gained</b> |
|------------------------------------------------------|----------------------|
| 12                                                   | 3 lbs                |
| 13                                                   | 4 lbs                |
| 14                                                   | 5 lbs                |
| 15                                                   | 6 lbs                |
| 16                                                   | 7 lbs                |
| 17                                                   | 8 lbs                |
| 18                                                   | 9 lbs                |
| 19                                                   | 10 lbs               |
| 20                                                   | 11 lbs               |
| <b>RULE: Difference of 9 → Week – 9 = lbs gained</b> |                      |

**Example:** Woman in her 28<sup>th</sup> week, she has gained 22 lbs. What is your impression?

- A) Fine
- B) Underweight
- C) Overweight
- D) Do an ax



Within 1-2 lbs = okay

3 lbs off = ax

4 lbs off = trouble

**Example:** If in 31<sup>st</sup> week gained 15 lbs → 31 – 9 is 22. Therefore, she is underweight by more than 4 lbs → do an ax like a biophysical profile.

### **FUNDAL HEIGHT**

**Fundus:** Top part of the uterus. Not palpable until week 12.

**1. Therefore, in 1<sup>st</sup> trimester you CANNOT palpate the fundus.**

What if she's gained 10 lbs in trimester, and fundus is WAY up high she's either not in the 1<sup>st</sup> trimester or she has hypermolar pregnancy (Cancerous).

**2. You can palpate the fundus after the END of the 1<sup>st</sup> trimester.**

**3. 2<sup>nd</sup> and 3<sup>rd</sup> Trimester → When is the fundus at the belly button/umbilicus?**

20 – 22 weeks of gestation.

**Why is this important for a nurse to know?**

You can use fundal height to ax the age of gestation. For instance, mom comes in ER after MVC. She is unable to tell you how far along she is.

**How can you put her in the correct trimester?**

Palpate her fundus.

Non palpable = 1<sup>st</sup> trimester. Mom is the priority.

At or below the belly button = 2<sup>nd</sup> trimester. Mom is the priority.

Above the belly button = 3<sup>rd</sup> trimester. Baby is the priority.

### **SIGNS OF PREGNANCY**

Only two categories on boards → POSITIVE and EVERYTHING ELSE.

#### **Four Positive Signs**

1. Fetal skeleton on XRAY
2. A fetal presence on U/S
3. Auscultation of fetal HR (8 – 12 weeks is when you can hear it although it starts beating at 5 weeks)

Most OB information has a range where it occurs because every woman is different. Because of that, be careful because there could be **3 different questions for every fact.**

**The first question would be:** When would you first auscultate a fetal heart? 8 wks

**The second question would be:** When would you most likely auscultate a fetal heart? 10 wks

**The third question would be:** When should you auscultate a fetal heart by? 12 wks

4. When the examiner palpates fetal movement NOT the mom

## **The MAYBE Signs (Probable & Presumptive together)**

1. All urine and blood pregnancy tests
  - A positive pregnancy test is NOT a positive sign of pregnancy = TRUE, because it means you have the hormonal increases but you may not have a fetus → false positive pregnancy test
2. Chadwick's, Goodell's, Hegar's
  - a. Chadwick's sign → cervical colour change to cyanosis (all those words start with C and so does Chadwick's). If the cervix is doing it, the vagina is too.
  - b. Goodell's sign → cervix softening
  - c. Hegar's sign → uterine softening
  - d. They occur in alphabetical order!!!
  - e. Boards tests more on the order not necessarily what week

## **PATIENT TEACHING IN PREGNANCY**

1. Good prenatal care → How often to come in for good prenatal care
  - a. Once a month until week 28 (1<sup>st</sup> trimester, 2<sup>nd</sup> trimester, up until 7<sup>th</sup> month)
  - b. AT week 28, once every two weeks until week 36
  - c. Then, every week until delivery OR week 42
    - i. At week 42 → C-section or induction

If woman comes in for 12<sup>th</sup> week check up, she comes in next at... 16<sup>th</sup> week. Then 20<sup>th</sup>, then 24<sup>th</sup>, then 28<sup>th</sup>, then 30<sup>th</sup>, then 32, 34, 36, 37, 38...

2. Hemoglobin will fall
  - a. Normal: 12-16
  - b. 1<sup>st</sup> trimester: 11 and it is normal.
  - c. 2<sup>nd</sup> trimester: 10.5 and still normal.
  - d. 3<sup>rd</sup> trimester: 10 and still normal.
3. Discomforts of pregnancy
  - a. Morning sickness
    - i. 1<sup>st</sup> trimester
    - ii. Tx: Dry carbohydrates before she gets out of bed
  - b. Urinary incontinence
    - i. 1<sup>st</sup> and 3<sup>rd</sup> trimester
    - ii. 2<sup>nd</sup> trimester → no incontinence because it's not on bladder it's in abdomen
    - iii. Tx: Void q2hrs all the way from the day she gest pregnant until 6 weeks after delivery
  - c. Difficulty breathing
    - i. 2<sup>nd</sup> and 3<sup>rd</sup> trimester
    - ii. Tx: Tripod position → Feet flat, arms on table, leaning forward
  - d. Back pain
    - i. 2<sup>nd</sup> and 3<sup>rd</sup> trimester
    - ii. Tx: Pelvic tilt exercises → tilt the pelvis forward

**That's it for pregnancy because the rest should be common sense → when you get a question you don't know the answer to → pick the answer that you would pick for someone else. Pregnancy is not a disease!**

## **LABOUR AND BIRTH**

### **TRUEST AND MOST VALID SIGN OF LABOUR**

- Onset of regular progressive contractions

### **TERMS**

- **Dilation:** Opening of cervix. It goes from 0-10cm. A cervix that is 0 = closed. A cervix that is 10 = fully dilated.
- **Effacement:** Thinning of the cervix. It goes from THICK to 100%. A cervix that is not effaced is described as thick. A completely effaced cervix is 100%.
- **Station:** Relationship of fetal presenting part to mom's ischial spines.
  - **Ischial Spines:** smallest diameter that the baby has to pass through in order to be birthed vaginally. If it cannot fit through there, it cannot be birthed vaginally.
  - **Negative station:** Presenting part is ABOVE ischial spines.
  - **Positive station:** Presenting part is BELOW ischial spines.
    - **Way to remember stations:** "Positive numbers are positive news. Negative numbers are negative news."
- **Engagement:** Station 0. Presenting part is AT the ischial spines.
- **Lie:** Relationship between spine of mother and spine of baby.
  - **Vertical lie:** Compatible with vaginal birth, uncomplicated.
    - Mom's spine and baby's spine are parallel = GOOD = BABY
  - **Transverse lie:** Baby's spine is perpendicular to mom's spine, complicated, trouble.
    - T which is transverse = BAD = TROUBLE
- **Presentation:** Part of baby that enters the birth canal first. Wonderful little alphabet soups.
  - Nobody really knows the presentations (it's not an average question)
  - Most common: ROA so just pick that one! Or LOA.

### **FOUR STAGES OF LABOUR AND DELIVERY**

1. **Labour – All of labour! Composed of THREE phases.**
  - a. **LATENT:** First three letters are "LAT" which are the initials of the phases in order!
  - b. **ACTIVE**
  - c. **TRANSITION**
2. **Delivery of baby**
3. **Delivery of placenta**
4. **Recovery**
  - a. How long does recovery last? 2 hours.

### **Questions:**

What is the purpose of uterine contractions in the 1<sup>st</sup> stage? To dilate and efface the cervix.

What is the purpose of uterine contractions in the 2<sup>nd</sup> stage? To push the baby out.

What is the purpose of uterine contractions in the 3<sup>rd</sup> stage? To push the placenta out.

What is the purpose of uterine contractions in the 4<sup>th</sup> stage? To contract the uterus to stop bleeding.

When does post partum technically begin? Two hours after the placenta is delivered.

What is the number one priority in the second PHASE? Pain management

What is the number one priority in the second STAGE? Airway of baby

Important nursing action in the third PHASE? Checking dilation, helping with pain, helping with breathing.

Important nursing action in the third STAGE? Making sure there's three vessels in the cord.

**Don't get mixed up between STAGES and PHASES.**